

Correction and Maturation of the CTMT Redshift Claim: From the Withdrawn “JILA Discrepancy Resolution” to a Conservative Tested-Class Reconstruction

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Abstract

This note replaces the stronger claim made in an earlier CTMT preprint (“Resolution of the JILA mm-scale redshift discrepancy”) with a narrower and more defensible statement. The earlier preprint incorrectly treated a static phase-jitter scale as if it were the accumulated gravitational phase drift over one second, thereby mixing a gauge/noise-like phase scale with the transport quantity relevant for redshift. That identification was wrong. The present paper explicitly withdraws that stronger wording and apologizes for the overstatement.

The corrected statement is more modest. In the weak-field regime relevant to the JILA millimetre-scale strontium clock experiment, the operational clock observable is the phase-drift rate

$$\dot{\Delta\varphi} = \bar{\omega} \delta, \quad \delta \approx \frac{gh}{c^2},$$

and the corresponding CTMT time functional is the coherence proper time

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\Delta\varphi}(t)}{\bar{\omega}} dt.$$

Under the tested admissibility conditions of Fisher identifiability, null-sector separation up to gauge, Čencov monotonicity under coarse-graining, inversion stability, and dimensional closure, this yields

$$\tau_{\text{coh}}(T) = \delta T$$

on the weak-field tested class. This is not presented as a universal derivation of general relativity, nor as a prediction of the JILA result. Rather, it is a conservative transport-class reconstruction statement: the JILA weak-field redshift and the CTMT synchrony-drift description lie in the same tested admissible class.

To clarify the maturity of the claim, we also report a synthetic adversarial battery attacking the identity $\tau_{\text{coh}}(T) = \delta T$ by controlled violations of linear transport, null-sector isolation, inversion stability, and Fisher sensitivity. The identity survives exactly in the baseline weak-field regime and fails quantitatively under each attack. This is consistent with the corrected interpretation that the corrected redshift reconstruction is not a tautological numerical fit but a structurally stable identity only within the intended admissibility regime.

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1 Introduction

This note has a corrective purpose. It replaces the earlier CTMT preprint claim entitled “Resolution of the JILA mm-scale redshift discrepancy” by a more conservative and more defensible statement.

The earlier wording was too strong. In particular, it suggested that a discrepancy with the JILA millimetre-scale gravitational redshift experiment had been *resolved* by CTMT. That is not the right description of what was established. The earlier argument mixed a static phase-jitter scale with an accumulated transport phase drift, and therefore conflated a gauge/noise-like local phase quantity with the physically relevant redshift observable.

The present note makes three things explicit:

- (I1) The author acknowledges that the stronger wording in the earlier preprint was an overstatement and apologizes for that mischaracterization.
- (I2) The specific mathematical error is identified and corrected.
- (I3) The mature claim is restated in a narrower form that is compatible with the current CTMT admissibility framework.

The central message of this corrected paper is deliberately modest:

In the weak-field millimetre-scale clock regime, CTMT does not derive general relativity and does not predict the JILA result from first principles. What CTMT can currently support is a tested-class reconstruction statement: when Fisher identifiability, Čencov monotonicity, inversion stability, null-sector separation, and dimensional closure hold, the coherence proper time reproduces the standard weak-field redshift relation

$$\tau_{\text{coh}}(T) = \delta T, \quad \delta \approx \frac{gh}{c^2}.$$

This is a transport-class statement, not a universal law.

2 Statement of correction and apology

Remark 1 (Correction of the earlier preprint). The earlier CTMT preprint entitled “Resolution of the JILA mm-scale redshift discrepancy” used a phase-scale comparison that was not operationally aligned with the experimental observable. In that draft, a static or local phase-jitter magnitude was effectively interpreted as if it were the accumulated phase drift induced by gravity over the measurement interval. This was an incorrect identification.

Remark 2 (Apology and replacement of claim). The stronger wording of “resolution” was not justified by the mathematics available at that stage. The author therefore withdraws that stronger formulation and apologizes for the overstatement. The present paper replaces it with a narrower, tested-class claim based on the corrected operational mapping and on explicit admissibility conditions.

This correction is not cosmetic. It marks a real maturation of the claim structure:

- from a *number-matching narrative*,
- to an *operational transport reconstruction*,
- with explicit *non-claims*, *failure modes*, and *falsifiability*.

3 Operational setting: weak-field redshift as a clock observable

We work in the ordinary weak-field setting of two clocks separated vertically by a small height difference h near the Earth’s surface. The standard gravitational redshift is

$$\delta_{\text{GR}} = \frac{\Delta\Phi}{c^2} \approx \frac{gh}{c^2}. \quad (3.1)$$

For the millimetre-scale setting

$$h = 1 \text{ mm}, \quad g = 9.80665 \text{ m/s}^2, \quad c = 2.99792458 \times 10^8 \text{ m/s},$$

one obtains

$$\delta_{\text{GR}} \approx 1.09 \times 10^{-19}. \quad (3.2)$$

In CTMT, the relevant observable is not a static phase offset but a *phase-drift rate*. Let $\bar{\omega}$ denote the local angular clock frequency. Then the operational transport relation is

$$\dot{\Delta\varphi} = \bar{\omega} \delta. \quad (3.3)$$

Integrating over a measurement interval $[0, T]$ yields

$$\Delta\varphi(T) = \bar{\omega} \delta T \quad \implies \quad \frac{\Delta\varphi(T)}{\bar{\omega}} = \delta T. \quad (3.4)$$

Thus, on the weak-field tested class, the per-second accumulated time offset is

$$\tau_{\text{coh}}(1 \text{ s}) = \delta. \quad (3.5)$$

For a strontium optical clock scale $\bar{\omega} \approx 2.7 \times 10^{15} \text{ rad/s}$, one obtains the phase-drift rate

$$\dot{\Delta\varphi} \approx (2.7 \times 10^{15})(1.09 \times 10^{-19}) \approx 2.95 \times 10^{-4} \text{ rad/s}. \quad (3.6)$$

That is the correct operational quantity to compare with CTMT synchrony drift.

4 What was wrong in the earlier estimate

The earlier discrepancy came from using the wrong object.

4.1 The incorrect identification

The earlier preprint effectively used a *static phase scale*

$$\Delta\varphi_{\text{static}} \sim 10^{-4} \text{ rad}$$

as though it were the *time-accumulated gravitational phase drift over one second*. This step is not operationally valid.

Static phase scale and phase-drift rate are different quantities:

- a static phase magnitude can represent noise, offset, calibration error, or gauge-like phase convention,
- a phase-drift rate is a transport quantity and carries the frequency-shift information relevant for redshift.

The weak-field redshift observable corresponds to

$$\dot{\Delta\varphi} = \bar{\omega} \delta,$$

not to an arbitrary local phase amplitude.

4.2 The mathematical source of the error

The earlier preprint therefore mixed two distinct sectors:

- (E1) a local phase/noise sector, which may contain gauge-like offsets or static jitter scales;
- (E2) a transport sector, which carries the accumulated phase drift and therefore the identifiable redshift.

From the mature CTMT perspective, the mistake can be stated as follows:

A local or static phase quantity was treated as if it were already the transport-invariant observable. It is not.

This is exactly the distinction that the mature CTMT formulation enforces: only the appropriate transport-integrated quantity survives the admissibility filters.

5 Mature CTMT formulation of proper time and redshift transfer

We now state the corrected CTMT objects.

5.1 Coherence proper time

Definition 1 (Coherence proper time). Let $\varphi(t)$ be the local clock phase and $\bar{\omega}$ the local transition angular frequency. The CTMT coherence proper time accumulated over the interval $[0, T]$ is

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\varphi}_{\text{phys}}(t)}{\bar{\omega}} dt, \quad (5.1)$$

where $\dot{\varphi}_{\text{phys}}$ denotes the admissible transport component of phase evolution after null-sector separation and operational filtering.

Remark 3. The point of (5.1) is that CTMT proper time is not coordinate time and not an arbitrary local phase convention. It is the normalized accumulation of the physically admissible phase-drift channel.

5.2 Why raw Fisher eigenvalue accumulation is not the definition

Earlier heuristic thinking may suggest a quantity such as

$$\tau(t) \sim \int \lambda_{\max}(F(t')) dt',$$

where F is a Fisher matrix. This is useful as an *identifiability heuristic*, but it is not by itself an admissible definition of CTMT proper time.

Remark 4. Raw eigenvalues of F are not transport-invariant in general and need not survive admissible reparametrization or operational normalization. The mature CTMT quantity is therefore (5.1), not a bare eigenvalue integral. Fisher enters as an admissibility and identifiability layer, not as a stand-alone clock functional.

5.3 Weak-field transport law

In the weak-field JILA regime, the admissible physical phase channel is

$$\dot{\varphi}_{\text{phys}}(t) = \bar{\omega} \delta,$$

so (5.1) reduces immediately to

$$\tau_{\text{coh}}(T) = \int_0^T \delta dt = \delta T. \quad (5.2)$$

Equation (5.2) is the transport-consistent weak-field realization obtained under the current CTMT admissibility constraints.

Its status is empirical and tested-class: it follows from the present inversion and transport pipeline but is not claimed to be universally valid beyond this regime.

6 Fisher, null sector, and Čencov structure

The corrected claim is not just the formula (5.2); it is the *admissibility context* under which the formula is allowed.

6.1 Fisher layer

Lemma 1 (Local Fisher identifiability of the redshift channel). *Let $\theta = \delta$ be the fractional redshift parameter, and suppose the clock phase is measured with effective Gaussian phase noise variance σ_φ^2 over an integration time T . Then, to leading order, the Fisher information for δ scales as*

$$F_{\delta\delta} \propto \frac{\bar{\omega}^2 T^2}{\sigma_\varphi^2}. \quad (6.1)$$

In particular, the weak-field redshift channel is non-null whenever the effective phase sensitivity is nonzero.

Remark 5. This means that the weak redshift is an identifiable direction. It is therefore not admissible to demote it to a pure gauge effect.

6.2 Null sector

Lemma 2 (Gauge/null sector). *Let ψ denote a global phase offset and let α denote a trivial global reparametrization of the time label that leaves the fractional shift δ unchanged. Then these directions belong to a gauge/null sector, whereas the redshift channel ∂_δ does not:*

$$\ker F \supset \text{span}\{\partial_\psi, \partial_\alpha\}, \quad \partial_\delta \notin \ker F. \quad (6.2)$$

Remark 6. This is exactly why the earlier use of a static phase scale was dangerous: phase-like local quantities may contain null/gauge contributions unless the transport channel is separated first.

6.3 Čencov monotonicity

Lemma 3 (Čencov monotonicity under admissible coarse-graining). *Let Π be an admissible coarse-graining of the weak-field clock record (for example temporal averaging or binning). Then the Fisher information in the redshift parameter satisfies*

$$F_\Pi(\delta) \leq F(\delta), \quad (6.3)$$

with equality only for sufficient statistics.

Remark 7. No admissible coarse-graining can create spurious sensitivity to δ . This is one reason the redshift channel remains structurally meaningful under the CTMT admissibility hierarchy.

7 Operational correction and tested-class interpretation

7.1 Operational correction

Lemma 4 (Corrected operational identification of the weak-field observable). *In the weak-field clock-comparison setting, the experimentally relevant redshift observable is the fractional frequency shift*

$$\delta \approx \frac{gh}{c^2},$$

equivalently encoded by the phase-drift rate

$$\dot{\Delta\varphi} = \bar{\omega} \delta.$$

It is not encoded by an arbitrary static phase magnitude or local phase-jitter scale.

Proof. The fractional redshift is defined operationally through a frequency difference between clocks at different gravitational potentials. A phase observable accumulates frequency information only through its time derivative. Therefore the relevant clock signal is the phase-drift rate $\dot{\Delta\varphi}$, not a static phase scale $\Delta\varphi_{\text{static}}$. Integrating the drift rate yields

$$\Delta\varphi(T) = \bar{\omega} \delta T,$$

hence

$$\frac{\Delta\varphi(T)}{\bar{\omega}} = \delta T.$$

This identifies the correct transport quantity and corrects the earlier mismatch. $\square$

Corollary 1 (Corrected coherence-time formula in the weak-field regime). *If the admissible physical phase channel is*

$$\dot{\varphi}_{\text{phys}}(t) = \bar{\omega} \delta,$$

then the coherence proper time

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\varphi}_{\text{phys}}(t)}{\bar{\omega}} dt$$

reduces to

$$\tau_{\text{coh}}(T) = \delta T.$$

Remark 8 (What has actually been established). [Lemma 4](#) and [Corollary 1](#) establish the corrected operational mapping and its immediate consequence for the weak-field coherence-time functional. This is the main technical correction relative to the earlier preprint.

7.2 Tested-class interpretation

Definition 2 (Weak-field tested class). The weak-field tested class consists of clock-comparison regimes in which:

- (A1) the redshift channel is Fisher-identifiable,
- (A2) null-sector directions are separated up to gauge,
- (A3) admissible coarse-graining satisfies Čencov monotonicity,
- (A4) inversion is stable within tolerance,
- (A5) dimensional closure and unit consistency hold.

Remark 9 (Tested-class CTMT interpretation). If both the weak-field GR description and the CTMT synchrony-drift description lie in the weak-field tested class, then [Corollary 1](#) can be interpreted as a tested-class compatibility statement: both descriptions assign the same operational value to the weak-field redshift channel through the coherence proper time.

Remark 10 (Why this is not a derivation theorem). [Remark 9](#) is not a theorem deriving GR from CTMT, nor a uniqueness theorem for all transport morphisms. It is an interpretation of the corrected operational formula inside a restricted admissibility class.

8 On circularity and scope

A natural objection is that the weak-field tested class is partly designed to preserve the very transport quantity that the paper then interprets. This objection is legitimate and should be stated explicitly.

The present paper therefore does not claim that the tested-class interpretation is a non-circular derivation theorem. The non-circular content lies earlier:

- (C1) the correction of the observable mismatch (static phase scale versus phase-drift rate),
- (C2) the definition of coherence proper time as a normalized transport accumulation,
- (C3) the explicit stratification of admissibility conditions under which that quantity is meaningful.

The tested-class interpretation is weaker. It says only that, once the operational observable has been corrected and once the admissibility conditions are made explicit, the weak-field GR and CTMT descriptions are compatible on that tested class. This is an interpretation claim, not a derivation theorem.

9 Explicit non-claims

To avoid any repetition of the earlier overstatement, the following non-claims are explicit.

- (N1) We do not claim that CTMT derives general relativity.
- (N2) We do not claim that CTMT predicts the JILA result independently of the weak-field redshift law.
- (N3) We do not claim universality beyond the weak-field tested class.
- (N4) We do not claim that the present argument resolves strong-field redshift, compact-object time dilation, or quantum gravity.
- (N5) We do not claim that coherence proper time has been uniquely forced across all conceivable morphism families.
- (N6) We do not claim that the synthetic adversarial battery is laboratory evidence.

These non-claims are not rhetorical modesty. They are part of the actual corrected claim structure.

10 How the claim matured

It is useful to summarize the maturation process explicitly.

Stage 1: early over-strong interpretation

The earlier preprint treated the phase layer too naively and conflated local phase scale with accumulated transport phase.

Stage 2: realization of operational mismatch

It became clear that the observable relevant for redshift is the rate

$$\dot{\Delta\varphi} = \bar{\omega} \delta,$$

not an arbitrary local phase offset or jitter amplitude.

Stage 3: admissibility stratification

The redshift channel was then placed inside the broader CTMT hierarchy:

- Fisher identifiability,
- null-sector separation,
- Čencov monotonicity,
- inversion stability,
- dimensional closure.

Stage 4: corrected proper time

The corresponding proper-time object was reformulated as the normalized transport accumulation

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\varphi}_{\text{phys}}(t)}{\bar{\omega}} dt.$$

Stage 5: adversarial stress

Finally, the identity

$$\tau_{\text{coh}}(T) = \delta T$$

was attacked synthetically by violating its weak points one by one. This step is important because it shows the corrected claim is not merely a rephrased fit, but a statement with explicit failure modes.

11 Sanity checks and failure-mode demonstrations

To support the corrected claim, we constructed a synthetic adversarial battery for the identity

$$\tau_{\text{coh}}(T) = \delta T.$$

11.1 Baseline constants

We used the weak-field JILA-scale numbers

$$h = 10^{-3} \text{ m}, \quad g = 9.80665 \text{ m/s}^2, \quad c = 2.99792458 \times 10^8 \text{ m/s},$$

so that

$$\delta = \frac{gh}{c^2} \approx 1.091137 \times 10^{-19}. \tag{11.1}$$

For the optical clock scale we used

$$\bar{\omega} = 2.7 \times 10^{15} \text{ rad/s},$$

giving the baseline phase-drift rate

$$\bar{\omega} \delta \approx 2.946070 \times 10^{-4} \text{ rad/s}. \quad (11.2)$$

11.2 Deviation observable

We measured failure of the identity through the relative deviation

$$\epsilon = \frac{\tau_{\text{coh}}(T)}{\delta T} - 1. \quad (11.3)$$

Thus:

$$\epsilon = 0 \iff \tau_{\text{coh}}(T) = \delta T.$$

11.3 Attack families

We attacked four distinct weak points.

Attack A: nonlinear transport distortion. We replaced the linear phase-drift law by

$$\dot{\varphi}(t) = \bar{\omega} \delta \left(1 + \beta \frac{t}{T} \right). \quad (11.4)$$

Then

$$\tau_{\text{coh}}(T) = \delta T \left(1 + \frac{\beta}{2} \right), \quad \epsilon = \frac{\beta}{2}. \quad (11.5)$$

Attack B: null-sector leakage. We injected an oscillatory phase-rate contamination:

$$\dot{\varphi}(t) = \bar{\omega} \delta + \nu \bar{\omega} \delta \sin(\Omega t), \quad \Omega = \frac{3\pi}{T}. \quad (11.6)$$

This yields

$$\tau_{\text{coh}}(T) = \delta T + \nu \delta \frac{1 - \cos(\Omega T)}{\Omega}, \quad \epsilon = \frac{2\nu}{3\pi}. \quad (11.7)$$

Attack C: inversion instability. We modeled recovery shrinkage by Tikhonov regularization:

$$r(\lambda) = \frac{\sigma_{\min}^2}{\sigma_{\min}^2 + \lambda}, \quad \tau_{\text{coh}}^{\text{rec}}(T) = r(\lambda) \delta T. \quad (11.8)$$

Thus

$$\epsilon = r(\lambda) - 1. \quad (11.9)$$

In our synthetic run we used

$$\sigma_{\min} = 1.64 \times 10^{-4}$$

as a representative small singular scale.

Attack D: Fisher attenuation with noise. We modeled attenuation of the observable redshift sensitivity by

$$\varphi(t) = a \bar{\omega} \delta t + \text{noise}, \quad (11.10)$$

with $a < 1$ and Gaussian phase noise. The redshift was then re-estimated by slope recovery in Monte Carlo. This is not an analytic proof but a synthetic numerical stress attack on the Fisher layer.

11.4 Representative numerical results

The representative $T = 1$ s attack summary is shown in Table 1.

Scenario	Attack strength	ϵ	Comment
baseline	0	0	Exact weak-field identity
nonlinear transport	$\beta = 0.2$	0.100000	10% overrun from transport nonlinearity
null leakage	$\nu = 0.2$	0.042441	4.24% bias from oscillatory contamination
inversion instability	$\lambda = 10^{-8}$	-0.271032	27.1% under-recovery from shrinkage
Fisher attenuation	$a = 0.2$	-0.800000	80% collapse of recovered redshift

Table 1: Representative synthetic adversarial attacks on the identity $\tau_{\text{coh}}(T) = \delta T$.

11.5 Monte Carlo Fisher attack

The Fisher attenuation Monte Carlo trend is shown in Table 2. The exact values depend on the synthetic noise model, but the trend is unambiguous: reducing sensitivity factor a drives increasingly severe negative bias in the recovered redshift.

a	F_{proxy}	$\text{mean}(\epsilon)$	$\text{RMSE}(\epsilon)$
1.00	7.290×10^{38}	1.62×10^{-3}	8.18×10^{-2}
0.70	3.572×10^{38}	-2.99×10^{-1}	3.10×10^{-1}
0.50	1.822×10^{38}	-5.01×10^{-1}	5.08×10^{-1}
0.20	2.916×10^{37}	-7.99×10^{-1}	8.03×10^{-1}
0.10	7.290×10^{36}	-8.99×10^{-1}	9.02×10^{-1}
0.05	1.822×10^{36}	-9.50×10^{-1}	9.53×10^{-1}
0.01	7.290×10^{34}	-9.89×10^{-1}	9.93×10^{-1}

Table 2: Synthetic Monte Carlo Fisher attenuation attack.

11.6 Interpretation of the battery

The battery is negative in the productive scientific sense: it shows how the corrected identity can be forced to fail.

Remark 11 (Purpose of the synthetic battery). The synthetic battery is not evidence for the physical truth of the identity

$$\tau_{\text{coh}}(T) = \delta T.$$

Its purpose is narrower: to demonstrate explicit failure modes when the assumptions behind the corrected weak-field formula are violated.

Remark 12 (Sanity-check outcome). In the synthetic baseline configuration, the corrected weak-field identity is recovered exactly. Under deliberately constructed violations of linear transport, null-sector isolation, inversion stability, or Fisher sensitivity, the relative deviation

$$\epsilon = \frac{\tau_{\text{coh}}(T)}{\delta T} - 1$$

departs from zero by the expected sign and scale.

Remark 13. This is not a theorem about Nature. It is a synthetic stress result showing that the corrected CTMT identity is not vacuous: it occupies a restricted admissibility regime and has explicit failure mechanisms outside that regime.

12 What this paper actually contributes

The contribution of this paper is fourfold.

(K1) It identifies and corrects the observable mismatch in the earlier preprint: a static phase-scale quantity was used where a phase-drift rate was required.

(K2) It reformulates the CTMT proper-time object as the coherence proper time

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\varphi}_{\text{phys}}(t)}{\bar{\omega}} dt.$$

(K3) It states explicitly the admissibility conditions under which the weak-field identity

$$\tau_{\text{coh}}(T) = \delta T$$

may be interpreted.

(K4) It provides sanity-check failure-mode demonstrations showing how the corrected identity breaks when those assumptions are deliberately violated.

This is a correction-and-maturation paper, not a universal derivation paper.

13 Falsification paths

The corrected claim is falsifiable in several directions.

Proposition 1 (Operational falsification). *If a weak-field clock-comparison regime satisfying Fisher identifiability, null-sector separation, Čencov monotonicity, inversion stability, and dimensional closure systematically fails the relation*

$$\tau_{\text{coh}}(T) = \delta T$$

beyond quantified uncertainty, then the present tested-class reconstruction claim must be weakened or rejected for that class.

Proposition 2 (Structural falsification). *If future physical or synthetic suites produce gauge-candidate scenarios for which the phase-drift channel is admissible yet null-leakage, inversion, or Fisher-attenuation attacks do not degrade the identity in the predicted direction, then the current decomposition of failure modes is incomplete and must be revised.*

Proposition 3 (Strong-field falsification). *If strong-field or highly nonlinear regimes are found in which an admissible CTMT proper-time reconstruction exists but (5.2) remains exactly linear without additional transport corrections, then the present expectation that nonlinearity, inversion loss, and null-sector mixing become relevant would also need to be reconsidered.*

14 Limitations

The scope of this correction paper is limited.

(L1) The argument is weak-field only.

(L2) The central claim is a tested-class reconstruction claim, not a universal transport theorem.

(L3) The synthetic attack battery is not laboratory evidence and does not substitute for experimental data analysis.

- (L4) The Fisher attenuation study is numerical and model-dependent.
- (L5) The inversion-instability attack uses a representative synthetic singular scale rather than a full laboratory forward model.
- (L6) The present note does not derive strong-field redshift corrections.
- (L7) The present note does not prove uniqueness of coherence proper time across all possible CTMT morphism families.

15 Conclusion

This paper corrects and replaces the earlier stronger claim of a “resolution” of the JILA millimetre-scale redshift discrepancy. The earlier preprint treated a static phase-like quantity as though it were the gravitationally accumulated transport phase over the measurement interval. That was the central mistake. The author acknowledges this error and withdraws the stronger wording.

The mature formulation is narrower but substantially more defensible. In the weak-field tested class, the appropriate CTMT clock observable is the phase-drift rate

$$\dot{\Delta\varphi} = \bar{\omega} \delta,$$

and the corresponding CTMT proper-time object is the coherence proper time

$$\tau_{\text{coh}}(T) = \int_0^T \frac{\dot{\varphi}_{\text{phys}}(t)}{\bar{\omega}} dt.$$

Under the tested admissibility conditions, this yields

$$\tau_{\text{coh}}(T) = \delta T, \quad \delta \approx \frac{gh}{c^2}.$$

This is not a derivation of GR and not a prediction of the JILA experiment. It is a transport-class reconstruction statement: the weak-field JILA redshift and the CTMT synchrony-drift description are operationally compatible within the same admissible tested class.

Finally, the synthetic adversarial battery shows that this corrected identity is not empty. It survives in the baseline weak-field regime and fails controllably when one violates linear transport, null-sector isolation, inversion stability, or Fisher sensitivity. That is the scientifically mature status of the claim.

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